



SEMESTER ONE EXAMINATIONS 2024/2025

MODULE: CSC1005 - Data Science & Databases

PROGRAMME(S):

DS	BSc Data Science
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1

EXAMINER(S):

Mark Roantree	(Internal)	(Ext: 5636)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

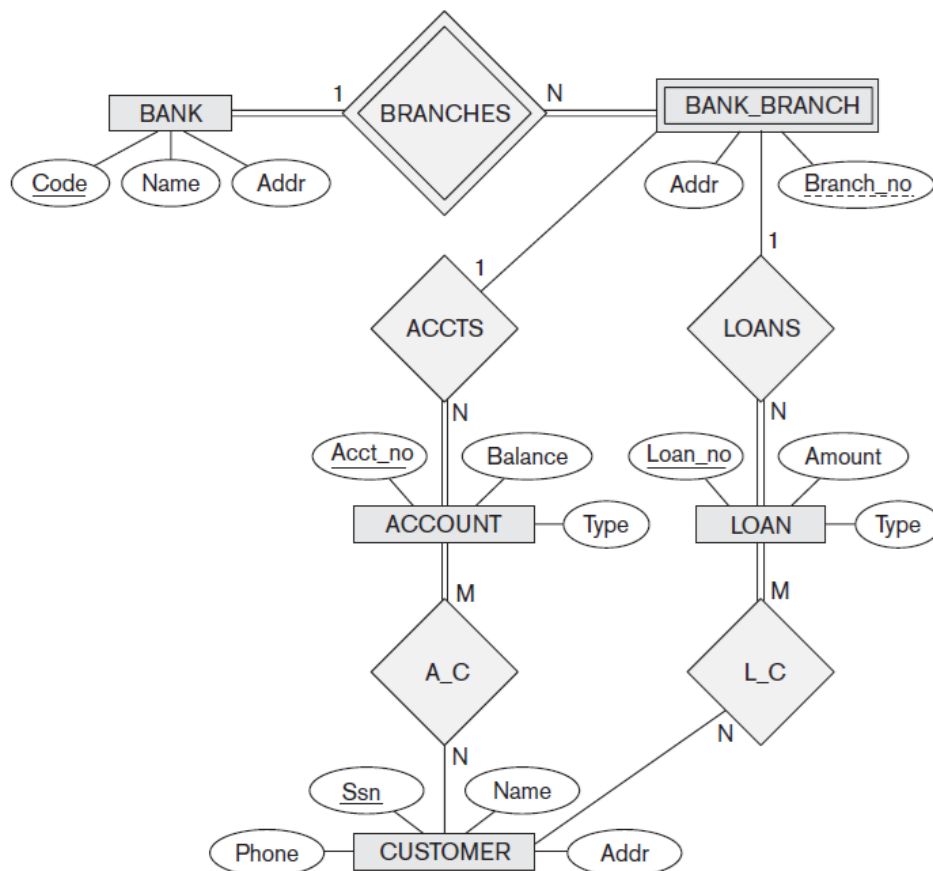
QUESTION 1 (Entity-Relationship Mapping)**[TOTAL MARKS: 25]**

Figure 1: Banking E-R Diagram

You are required to use the Entity Relationship (E-R) Mapping process to convert the above E-R diagram to a database schema. The first 6 steps form the basis of the question parts (a) to (f) below. Use 2 separate diagrams: the first to answer parts a) to c) and create a separate diagram for parts d) to e).

Q 1(a) [4 Marks]

What is the goal of first step in mapping an E-R diagram to a relational database? Show the result of this first step using figure 1.

Q 1(b) [6 Marks]

What is the goal of second step in mapping an E-R diagram to a relational database? Update the schema from part (a) but make it clear which part of the schema is the answer to question (b).

What type of participation constraint is present?

Q 1(c) [1 Marks]

Why does nothing happen for the next (3rd) step in the process?

Q 1(d) [6 Marks]

What happens in Step 4 of the process? Show the updated schema **in a new diagram**. Explain the decision you made with respect to foreign key placement.

Q 1(e) [6 Marks]

What happens in Step 5 of the process? Show the updated schema **and make clear what has happened in this step**. Make sure that the primary keys for all tables are clearly identified.

Q 1(f) [2 Marks]

Finally, your 6th step should focus on multi-valued attributes. What takes place at this step using figure 1?

[End of Question 1]

QUESTION 2 (Relational Model)

[TOTAL MARKS: 25]

VID	Fname	Lname	Booth	CID	ProductID	Quantity	VID	Date
A01	Jim	Jones	B1	C01	P1	5	A01	10/10/2024
A02	Una	Smyth	B2	C02	P2	10	A01	10/10/2024
A03	Mary	Murphy	B3	C03	P3	2	A04	11/10/2024
A04	Valerio	Conte	B4	C04	P4	12	A03	11/10/2024
				C05	P5	8	A03	11/10/2024

Vendor Table

Customer Table

Figure 2: SALES schema with Vendor and Customer Tables.

Q 2(a) [4 Marks]

What is the intension of the SALES Schema?
What is the degree of the Customer Table?

Q 2(b) [6 Marks]

Compute the cartesian product of Vendor and Customer. Write the **entire result set** into your answer sheet.

Q 2(c) [6 Marks]

Perform an INSERT operation to Customer with the following values (C06, P6, 6, A06, 6/10/2024). What constraint is violated by this INSERT operation? Explain your answer.

Q 2(d) [9 Marks]

What other types of constraints can be violated through an INSERT operation on the Customer table. Name each type of violation and provide an example for each one.

[End of Question 2]

QUESTION 3 (Functional Dependencies)

[TOTAL MARKS: 25]

This question uses the STUDENTS and PROFESSORS tables in the Appendix.

Q 3(a)

[10 Marks]

Assuming that a functional dependency is of the form $L \rightarrow R$ where L represents one or more (determinant) attributes and R represents a different set of attributes, list 5 functional dependencies for the STUDENTS relation. Provide a **different value** for L in each dependency.

Q 3(b)

[6 Marks]

What is meant by the *dependency preservation* property? In your answer, explain the circumstances in which it may occur and how it should be resolved.

Q 3(c)

[9 Marks]

Assume you have a relation R with attributes (a,b,c,d,e). Briefly describe an algorithm to locate all functional dependencies in R .

[End of Question 3]

QUESTION 4 (Normalisation)

[TOTAL MARKS: 25]

This question uses the STUDENTS and PROFESSORS tables in the Appendix.

Q 4(a)

[6 Marks]

- I. In what normal form would you consider the STUDENTS table to be? Why?
- II. In what normal form would you consider the PROFESSORS table to be? Why?

Q 4(b)

[13 Marks]

Taking the relation STUDENTS, explain the steps you would take to bring this relation to third normal form. At each step, highlight the specific functional dependency that informed your decision for decomposition.

Q 4(c)

[6 Marks]

Take the PROFESSORS table and write an un-normalised version of this table to your answer book. Explain what you did and why it became an un-normalised table.

[End of Question 4]

QUESTION 5 (SQL Programming)**[TOTAL MARKS: 25]****Q 5(a)****[4 Marks]**

Write a query that determines how many times each vendor has rented a booth at the farmer's market.

Q 5(b)**[5 Marks]**

Which vendor did Bonnie Hassan spend most money with?

Q 5(c)**[8 Marks]**

Write a query to display the product category name, product name, earliest date purchased, and latest date purchased for every product in the "Fresh Fruits & Vegetables" product category:

Q5 (d)**[8 marks]**

Create a list of customers who have spent more than \$50, sorted by last name, then first name.

[End of Question 5]**Appendix**

SID	Name	Address	CourseID	CourseName	TutorID	TutorName
20181011	Michael Murphy	Dundalk	DC101	Computer Applications	750001	Jack Sharp
20181015	Ann Burke	Dunboyne	DC101	Computer Applications	750003	Pauline Burke
20181022	Tom Flynn	Finglas	DC101	Computer Applications	750004	Nora Martin
20181111	Mary Burke	Glasnevin	DC101	Computer Applications	750004	Nora Martin
20181211	Brian Byrne	Glasnevin	DC102	Enterprise Computing	750004	Nora Martin
20181256	Ellen Molloy	Glasnevin	DC102	Enterprise Computing	750007	Eilis Flynn
20181260	Deirdre flynn	Finglas	DC101	Computer Applications	750007	Eilis Flynn
20181275	Alan Marr	Finglas	DC102	Enterprise Computing	750001	Jack Sharp

Table A: STUDENTS relation

CourseID	Department	Head
DC101	Department of Computing	Prof. Murphy
DC102	Department of Biology	Prof. Swan
DC103	Department of Physics	Prof. McDonald

Table B: PROFESSORS relation**[END OF EXAM]**